

Blending Mixed Reality and Generative AI to Teach Geography: An MR+GenAI Learning Environment

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This study examines “VirtualGeo”, a Mixed Reality and Generative AI platform designed to enhance U.S. geography knowledge among international students. By integrating immersive technologies, VirtualGeo allows students to engage with spatial content within an interactive digital landscape. Using a mixed-methods approach, the study evaluated the platform’s impact through pre- and post-tests measuring geographic knowledge, map-drawing skills, and cognitive load,

complemented by qualitative interviews to capture students' learning experiences. Findings show significant gains in international students' geographic understanding and underscore VirtualGeo's potential to support adaptive and student-centered learning. Implications suggest that VirtualGeo is a scalable tool to teach geography and other spatial disciplines, which highlights new possibilities for research, policy, and practice in digital education.

INTRODUCTION

As of the 2023/24 academic year, 1,126,690 international students were enrolled in U.S. institutions, contributing significantly to campuses' economic, cultural, and academic vitality (Institute of International Education, 2024). However, limited prior exposure to U.S. geography often hinders their understanding of regional contexts and cultural nuances, making it challenging to navigate new environments and fully engage in academic and social settings (Great Learning Editorial Team, 2023). Mastering U.S. geography fosters spatial thinking, cultural integration, and ethical decision-making, enhancing students' understanding of physical and cultural landscapes that shape global dynamics (Solem & Weiguo, 2018). This knowledge aids navigation, supports engagement with local and global issues, and promotes cultural awareness and global citizenship—key for fostering international peace and understanding (Lidstone & Stoltman, 2002; Niu et al., 2023).

Emerging technologies like mixed reality (MR), virtual reality (VR), and generative AI (GenAI) offer significant potential to address these challenges through immersive, personalized learning environments. These tools can help international students better understand U.S. geography while improving their spatial awareness and cultural familiarity. However, as the application of GenAI is relatively recent, its impact on learning outcomes and experiences—particularly in combination with MR—remains unclear.

This study explores VirtualGeo, an MR+GenAI platform for U.S. geography learning. It examines learners' experiences and outcomes in this integrated context, offering insights into how advanced technologies can help bridge geographic and cultural knowledge gaps.

LITERATURE REVIEW

Immersive Technologies in Geography Education

In geography education, MR/VR technologies create immersive 3D environments, enabling students to explore landscapes and spatial concepts interactively (Bos et al., 2022). This is especially valuable for international students, who often lack both first-hand exposure to and direct instruction in U.S. geography. Cultural and geographical barriers significantly shape international students' geographic learning experiences and adaptation to new environments (Perez-Encinas & Rodriguez-Pomeda, 2019). Moreover, time and financial constraints limit their ability to explore and familiarize themselves with their surroundings, compounding challenges in navigating and integrating into academic and social communities (Gokpinar-Shelton & Pike, 2022). This restricted engagement can lead to feelings of isolation and hinder their ability to connect with local communities.

MR/VR's immersive capabilities support deeper, more intuitive comprehension of spatial relationships (Macrine & Fugate, 2022), offering accessible and engaging ways to bridge these gaps. By providing immersive and interactive environments, these technologies help international students build geographic knowledge and spatial understanding, promoting their academic and social integration (Slater & Sanchez-Vives, 2016a; Xu et al., 2022).

While VR immerses users in fully virtual environments for experiential learning (Slater & Sanchez-Vives, 2016a), MR blends virtual and augmented reality to create interactive, hands-on experiences that enhance learning while keeping learners connected to their physical surroundings (Chatain et al., 2023; Johnson-Glenberg et al., 2020). MR is particularly effective for geography education, which benefits from contextualized, spatially grounded learning—an approach aligned with VirtualGeo design objectives.

GenAI in Personalized Learning

Artificial intelligence (AI) has emerged as a transformative tool in education, offering adaptive and personalized learning experiences that cater to individual needs. By leveraging machine learning algorithms and natural language processing, AI can analyze learners' performance and provide tailored feedback in real-time, enhancing engagement and improving educational outcomes. GenAI, a subset of AI, further extends these capabilities by

generating customized content, simulating real-world scenarios, and offering interactive problem-solving opportunities (Dogan et al., 2023; Zawacki-Richter et al., 2019).

In geography education, GenAI has the potential to bridge gaps in geographic literacy by creating realistic simulations of environments and providing learners with guided exploration of spatial concepts (Alshahrani, 2023). GenAI adjusts content to individual needs, offers real-time feedback, and customizes learning paths, supporting a student-centered approach that boosts motivation and educational outcomes (Maghsudi et al., 2021; Zawacki-Richter et al., 2019). These capabilities align with broader trends in education that emphasize personalized learning as a pathway to better comprehension and retention (Ng & Fung, 2024). Moreover, GenAI facilitates the development of dynamic, simulation-based learning environments that encourage exploratory learning and real-world problem-solving (Borge et al., 2024; Ryan Idea Lab, 2024).

Immersive Technologies and GenAI for Learning Innovations

The integration of immersive technologies like VR and MR with GenAI is transforming educational environments, creating opportunities to foster active learning, critical thinking, and problem-solving across disciplines (Alhafis Hashim et al., 2024). Immersive technologies bridge real and virtual elements, creating interactive and context-rich experiences, while GenAI enhances adaptability by generating personalized content and simulating real-world challenges, aligning the learning environment with individual needs.

For example, Doswell et al. (2020) illustrated how AI-enhanced virtual platforms empower learners to safely navigate high-stakes scenarios, providing real-time feedback and skill assessments that promote continual improvement. Additionally, Chheang et al. (2024) highlighted the impact of GenAI-powered simulations in medical education, where adaptive learning paths and interactive problem-solving exercises significantly enhance knowledge retention and learner engagement.

Summary and Research Questions

Existing research highlights the transformative potential of integrating immersive technologies with GenAI to create interactive, adaptive, and con-

text-rich learning environments. However, significant gaps remain in understanding their application for international students, particularly in addressing challenges related to limited exposure to U.S. geography and adaptation to local cultural and geographic contexts.

This study seeks to address these gaps by focusing on the combined use of MR and GenAI within the VirtualGeo platform to support U.S. geography learning for international students. By employing an embodied learning approach, VirtualGeo empowers students to interact with geographic concepts actively—drawing state shapes, identifying cities, locating rivers and mountains, and engaging with GenAI to explore geographic details. The study is guided by four research questions (RQs):

RQ1: To what extent does VirtualGeo enhance international students' knowledge of U.S. geography?

RQ2: How do international students interpret their embodied experience and cognitive load while participating in the VirtualGeo intervention?

RQ3: What are international students' learning experiences of using VirtualGeo?

RQ4: How does VirtualGeo contribute to international students' understanding of U.S. geography?

METHODOLOGY

In this study, we implemented a mixed-methods approach to evaluate the effectiveness of VirtualGeo, aiming to gain comprehensive insights into students' learning experiences, knowledge gains, map drawing skill improvement, the relationship between cognitive load and embodied experiences, and overall learning perceptions. We received the IRB approval from our institution before the study was implemented.

Learning System Design

VirtualGeo worked on Meta Quest Pro, an MR headset, with two controllers and a painting application, which was powered by ChatGPT, a GenAI system. Participants, while wearing the MR headset and holding onto controllers, used the painting application to interact with the learning content displayed on a TV screen (**Figure 1**). Participants would trace the out-

line of a U.S. state based on a provided map and mark the capital and major cities. Furthermore, they would illustrate rivers and mountains on a full U.S. map and orally talk to ChatGPT to learn more about a specific state or the U.S. as a whole. This way, participants had a personalized learning path by engaging with the learning content and exploring knowledge out of their own curiosity.

Figure 1

A participant is using VirtualGeo to trace the outline of the state of Pennsylvania



Research Design

Participants

For the exploratory nature of the study, seven first-year international students (**Table 1**) from a midwestern land-grant university in the United States were recruited, with 57% male and 43% female. Regarding familiarity with U.S. geography, 43% were not familiar at all, 43% were slightly familiar, and 14% were moderately familiar. Additionally, 43% had prior experience with virtual reality and/or mixed reality, while 57% had no such experience.

Table 1

Demographic characteristics of the participants (n=7)

Characteristics	Category	Participants Sample	
		n	%
Gender	Male	4	57%
	Female	3	43%
Age	24-28	3	42%
	29-33	2	29%
	34-38	2	8%
Familiarity with U.S. Geography	Not familiar at all	3	43%
	Slightly familiar	3	43%
	Moderately familiar	1	14%
	Very familiar	0	0%
Experience with Virtual Reality	Yes	3	43%
	No	4	57%

Instruments

The knowledge test aimed to assess the impact of VirtualGeo on participants’ U.S. geography knowledge, specifically focusing on map drawing skills and geographic comprehension. The knowledge assessment comprised 11 multiple-choice questions, three matching questions, and one map drawing task that evaluated participants’ ability to identify and draw U.S. states, cities, expressways, rivers, and mountains. The test items were verified and validated by a subject expert in geography.

Survey questions assessing embodied learning experiences were adapted from Lewis and Lloyd (2010) and Asai et al. (2016), tailored to align with VirtualGeo’s MR+GenAI context. Cognitive load was measured using items adapted from Chertoff et al. (2010), Hart & Field (2006), and related studies, customized to reflect the specific demands of the VirtualGeo intervention. All items were rated on a 5-point Likert scale.

The semi-structured interview questions were developed through a consensus reached by the research group after multiple discussions. These questions primarily explored the participants’ subjective impressions of the intervention. Examples include: “Do you think the use of ChatGPT motivates you to learn about this content?” and “Did you experience any challenges or frustrations?”

Study procedure and data collection

The data collection took place in a classroom with ample space, allowing participants to move and draw with their hands as much as they wanted (**Figure 2**). Each participant first signed a consent form, knowing their rights during the data collection. Then, they completed a pre-survey, which inquired about their demographic information and previous experience with VR/MR. The pre-survey contained the knowledge test. After the pre-survey, participants interacted with the MR+GenAI system for about 20 minutes. Upon their completion of the experience, they completed a post-survey which measured their embodied learning experiences and cognitive load. Similar to the pre-survey, the post-survey contained the same knowledge test. A follow-up interview was conducted with each participant immediately after the post-survey.

Figure 2

A participant uses VirtualGeo to learn US geographic knowledge immersively.



Data Analysis

Five graders assessed the map drawing task using standardized criteria. Knowledge gains and map drawing improvement were examined by comparing the paired pre- and post-test means, with paired sample *t*-tests conducted to examine any statistically significant differences in scores. In addition, we conducted Pearson and Spearman correlation analyses to investigate any significant correlations among participants' embodied learning experiences and their cognitive load.

An inductive thematic analysis approach (Braun & Clarke, 2012) was employed to identify and interpret patterns within the interview data, providing an in-depth understanding of the underlying meanings and themes. This flexible method is particularly suitable for exploratory research, as it facilitates a deeper exploration of complex phenomena within specific contexts. In this study, thematic analysis was used to capture students' perceptions of how the VirtualGeo experience influenced their understanding of U.S. geography and their overall learning experiences.

RESULTS

Quantitative Results

To measure geographic knowledge, paired sample *t*-tests were conducted on students' ability to recognize and locate U.S. states, cities, expressways, rivers, and mountains. The mean pre-test score was 10.9 out of 100 ($SD = 19.6$), which increased to 80.7 ($SD = 12.6$) in the post-test, showing a significant improvement, $t(6) = -7.50$, $p < .001$. This sharp increase underscores VirtualGeo's potential for bolstering knowledge acquisition in geography.

In map drawing skills, students demonstrated notable advancements as well. The mean pre-test score for this task was 4.29 out of 10 ($SD = 2.76$), which rose to 7.21 ($SD = 0.93$) in the post-test, yielding a statistically significant difference, $t(6) = -2.76$, $p = .033$. These results, illustrated in **Figures 3 and 4**, indicate that the immersive, interactive experience provided by VirtualGeo contributed meaningfully to students' ability to accurately trace and identify geographic features.

Figure 3

Improvement of U.S. geographic understanding between pre- and post-test

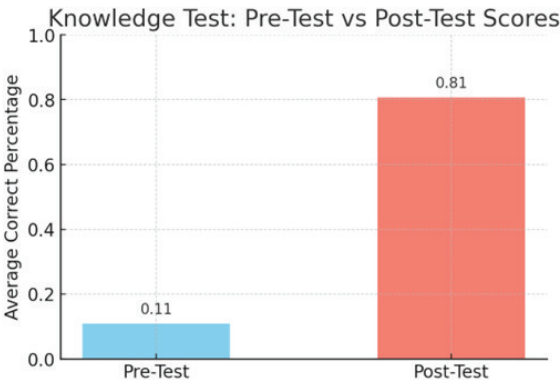
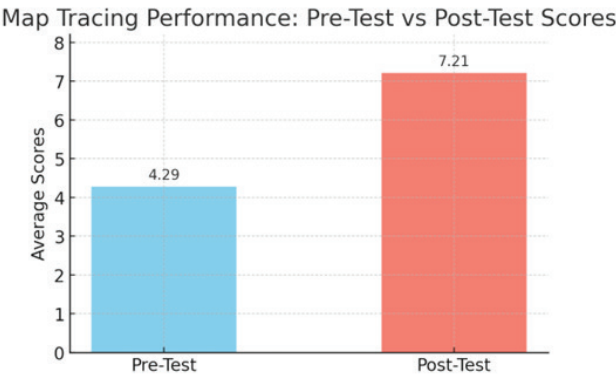


Figure 4

Improvement of map tracing skills between pre- and post-test



In addition to knowledge gains, post-survey data on embodied experience and cognitive load were analyzed. Participants provided positive evaluations of their embodied experience, as reflected by an overall mean score of 3.71 ($SD = 0.32$) on a scale of 1 to 5, highlighting the immersive quality of the learning environment. Additionally, cognitive load was assessed with

an overall mean score of 2.72 ($SD = 0.73$) on a scale of 1 to 5, suggesting a moderate level of mental and physical demand experienced by participants during the task. To further explore the relationship between the two constructs, Pearson and Spearman correlation analyses were conducted. The results indicated a moderately strong positive correlation between the two factors (Pearson $r = 0.697$; Spearman $r = 0.717$), suggesting that the physical engagement within VirtualGeo enhances cognitive engagement without overloading cognitive resources.

Qualitative Results

The interviews in this study revealed three main themes: engaging and immersive learning, educational benefits of MR and GenAI, and usability and realism challenges (Table 2).

Table 2

Summary of themes from the qualitative analysis of the VirtualGeo experience

Themes	Sub-Themes	Sample Codes
Engaging and Immersive Learning	Active Engagement	<i>“Drawing in VR was really active learning.”</i>
		<i>“It felt like exploring rather than studying.”</i>
		<i>“I was fully absorbed and forgot I was in a classroom setting.”</i>
	Novelty and Fun	<i>“This was my first time using VR, and it was exciting.”</i>
		<i>“The experience made geography fun and less like a lecture.”</i>
		<i>“Learning felt more like a game.”</i>
	Physical Involvement	<i>“Using my hands to draw helped me remember the geography better.”</i>
		<i>“The movement added a hands-on element that made learning more engaging.”</i>
		<i>“It felt like I was truly mapping things out myself.”</i>

Themes	Sub-Themes	Sample Codes
Educational Benefits of MR and GenAI	Independence and Motivation	<i>“The AI allows me to ask questions and learn independently.”</i> <i>“It’s motivating to explore when I get instant feedback.”</i> <i>“I didn’t need to rely on a teacher constantly.”</i>
	Enhanced Understanding	<i>“ChatGPT explained complex concepts in simple language, making it easier to understand.”</i> <i>“I remembered details better after using the VR environment.”</i> <i>“It was like having a tutor available all the time.”</i>
	Real-Time Feedback	<i>“ChatGPT answers were immediate, helping me follow along without needing a teacher.”</i> <i>“I could check my understanding quickly by asking ChatGPT.”</i> <i>“I liked that I could immediately clarify points I didn’t understand.”</i>
Usability and Realism Challenges	Usability and Visual Accuracy	<i>“Some eye strain with long use.”</i> <i>“It was hard to make sure my lines on the map were in the right place.”</i> <i>“I struggled with the headset view getting blurry.”</i>
	Realism and Accuracy in VR	<i>“It didn’t feel exactly like the real landscape, especially with complex features.”</i> <i>“The virtual maps were a bit blurry, making it hard to see details clearly.”</i> <i>“The maps weren’t as detailed as I expected for learning.”</i>
	Need for Additional Guidance	<i>“Having a short tutorial at the start would help with the controls.”</i> <i>“I felt lost at first, but I adjusted after some guidance.”</i> <i>“I wished there were more tips on using VR for beginners.”</i>

Engaging and immersive learning

Six out of seven participants found VirtualGeo to be an enjoyable and innovative approach to learning U.S. geography. The immersive VR environment made learning interactive rather than passive, with activities like drawing map elements enhancing their memory of geographic features. Comments included, *“It was really active learning... I really enjoyed drawing the lines of the states,”* and *“the whole experience felt like exploring rather than just studying.”*

Educational benefits of MR and GenAI

ChatGPT integration enabled participants to ask questions and get instant feedback, promoting independent learning. Five out of seven participants felt the AI's immediate responses increased engagement, with remarks such as, *“ChatGPT provides immediate answers, which helps me learn more as I explore,”* and *“The AI combined with VR allows me to learn independently without needing an instructor.”*

Usability and realism challenges

Four out of seven participants experienced VR-related issues, including blurriness and eye strain during longer sessions. For example, one participant remarked, *“After a while, my eyes felt strained from the headset, especially during extended use,”* while another noted, *“The sight was quite blurred, which made it hard to focus on the map details.”* They also pointed out that the VR environment lacked full realism for more complex landscapes. One participant explained, *“It didn't feel exactly like the real landscape, especially for rivers and mountains,”* and another commented, *“When I was drawing the map, I thought I was accurate, but the final lines didn't match my intent.”*

DISCUSSION AND CONCLUSION

This study evaluated VirtualGeo, an MR and GenAI platform, showcasing its potential to transform U.S. geography education for international students through interactive, immersive, embodied experiences. The plat-

form's design promotes engagement with complex spatial concepts, such as map orientation and geographic features, aligning with existing research on MR's effectiveness in facilitating embodied learning and cognitive engagement (Chatain et al., 2023; Johnson-Glenberg et al., 2020).

The first research question explored how well VirtualGeo enhances international students' knowledge of U.S. geography. The results demonstrated significant gains, with over a 60% improvement in knowledge scores and notable advancements in map-drawing accuracy. By providing an immersive and interactive environment, the platform addresses common challenges faced by international students, particularly those unfamiliar with U.S. geography, by making spatial relationships more tangible and engaging.

The second research question examined how participants interpreted their embodied experience and cognitive load during the VirtualGeo intervention. Correlation analyses indicated that physical interaction within the MR environment enhanced cognitive focus while avoiding cognitive overload. This aligns with theories of embodied cognition, which suggest that physical engagement deepens cognitive connections and improves the retention of complex spatial information (Johnson-Glenberg, 2018; Weisberg & Newcombe, 2017). Activities like tracing state boundaries and exploring virtual landscapes allowed participants to connect their physical actions with the digital content, reinforcing learning in a meaningful way.

Qualitative findings further emphasized VirtualGeo's value as an interactive and transformative learning tool. Participants described the platform as enjoyable and highly immersive, noting that tasks such as drawing geographic features and interacting with ChatGPT enhanced their comprehension and retention. Participants likened the AI to a supportive tutor or peer, underscoring its role in creating a dynamic, learner-centered experience. These insights align with research highlighting the motivational benefits of personalized, adaptive learning environments (Maghsudi et al., 2021; Zawacki-Richter et al., 2019).

The third and fourth research questions explored participants' overall learning experiences and how VirtualGeo contributed to their understanding of U.S. geography. The findings revealed that the platform effectively transformed traditional, passive learning into an active, exploratory process. Participants highlighted how the MR+GenAI approach allowed them to visualize spatial relationships more intuitively, making abstract geographic concepts easier to grasp.

In addition to advancing students' knowledge and engagement, VirtualGeo offers a scalable model for integrating MR and GenAI into diverse learning environments. By offering interactive experiences accessible anytime as needed, the platform bridges self-paced learning and instructor-led

education. AI-driven feedback empowers independent learning, reducing reliance on instructors while addressing individual needs (Alshahrani, 2023). These features position VirtualGeo as a practical solution for diverse educational contexts, aligning with goals to democratize access to high-quality, interactive learning resources (Potkonjak et al., 2016; Slater & Sanchez-Vives, 2016b). Its ability to foster spatial reasoning and critical thinking also extends to STEM and global citizenship education.

In conclusion, VirtualGeo demonstrates the transformative potential of MR and GenAI, offering an innovative way to teach complex subjects like geography. By fostering engagement, autonomy, and critical thinking, it paves the way for broader applications in interdisciplinary education, preparing learners for an interconnected world.

LIMITATIONS AND FUTURE DIRECTIONS

This study highlights VirtualGeo's potential to improve U.S. geography knowledge for international students but notes limitations. A small sample size limits generalizability, and self-reported data may introduce bias. Usability issues, such as discomfort and eye strain, could affect engagement, and long-term retention was not assessed. Future research should include larger and more diverse samples, investigate long-term impacts, and refine the MR setup to improve comfort and accessibility. Moreover, developing mobile-friendly AR versions could increase accessibility in under-resourced settings. As part of a new research initiative, we plan to include more American students, particularly those with limited travel experience, to broaden the scope and applicability of the findings.

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